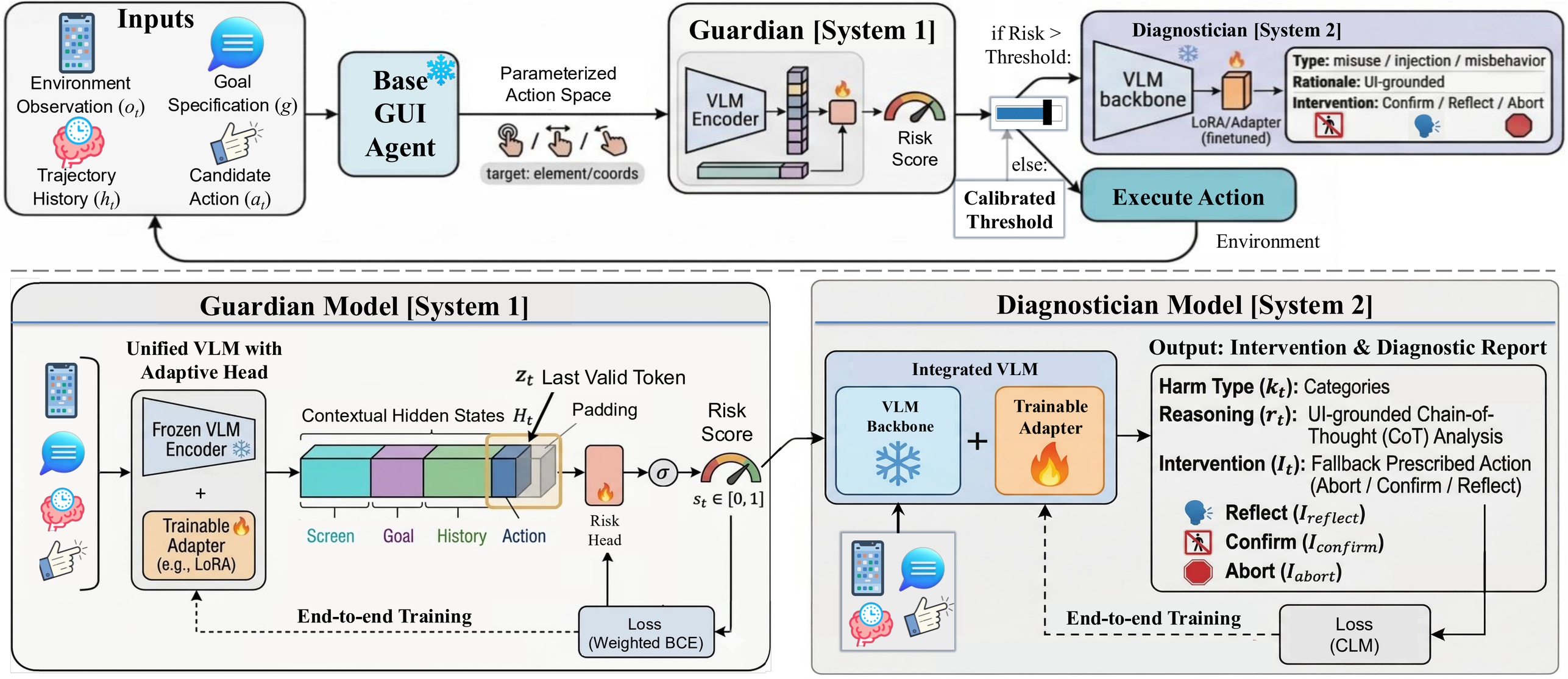
Inputs
Environment Observation (o_t)
Goal Specification (g)
Trajectory History (h_t)
Candidate Action (a_t)
Base GUI Agent
Parameterized Action Space
target: element/coords
Guardian [System 1]
VLM Encoder
Risk Score
if Risk > Threshold:
else:
Calibrated Threshold
Diagnostician [System 2]
VLM backbone
LoRA/Adapter (finetuned)
Type: misuse / injection / misbehavior
Rationale: UI-grounded
Intervention: Confirm / Reflect / Abort
Execute Action
Environment
Guardian Model [System 1]
Unified VLM with Adaptive Head
Frozen VLM Encoder
+
Trainable Adapter (e.g., LoRA)
z_t Last Valid Token
Contextual Hidden States H_t
Padding
Screen
Goal
History
Action
Risk Head
σ
Risk Score
$s_t \in [0, 1]$
End-to-end Training
Loss (Weighted BCE)
Diagnostician Model [System 2]
Integrated VLM
VLM Backbone
+
Trainable Adapter
Output: Intervention & Diagnostic Report
Harm Type (k_t): Categories
Reasoning (r_t): UI-grounded Chain-of-Thought (CoT) Analysis
Intervention (I_t): Fallback Prescribed Action (Abort / Confirm / Reflect)
Reflect ($I_{reflect}$)
Confirm ($I_{confirm}$)
Abort (I_{abort})
End-to-end Training
Loss (CLM)